MD Simulation and Homology Modelling

Basic principles and applications

(November 2020)

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Preface

Expectation vs reality, and the meaning of doing research

Your expectations?

1. What kind of “research experience” you are looking for?
2. What you expect a computer can do?

Reality

1. What kind of “research experience” you are looking for?
   - Research =/= achievement
   - Academia =/= let you hide from the job market
2. What you expect a computer can do?
   - Main advantages:
     - Virtually screen and design inhibitors → reduce experimental cost
     - Study phenomena that are difficult to be investigate experimentally
   - You think its easier to handle than experimental work?
     - NO! the learning curve can be much steeper
   - **LIMITS: Computer cannot do everything**
     - →**Still have false positive/false negative hits!!!!!**
     - →**Possible to get problematic results due to the limits of a certain method!!!!!**
     - →**Note the assumptions you are making!!!!!!**
     - →**Always try to validate your results if possible!!!!!**
     - →**Be VERY careful with the predicted/generated results!!!!!!**

What does it mean by doing research?

- Working according to the manual?
- Strictly follow your supervisor's instruction?
- 100% sure to have some interesting results for your presentation?

x **Researcher**
x **Scientist**
✓ **User of an instrument/a software**
✓ **Worker**

What does it mean by doing research?

- ~~Working according to the manual?~~
  →Know how to work according to a manual + how to work without a manual

- ~~Strictly follow your supervisor's instruction?~~
  →Understand the reason and suggest improvements if possible

- ~~100% sure to have some interesting results for your presentation?~~
  →Learn from failures / undesirable results
  →Accept the fate when there aren't any promising results

What does it mean by doing research?

1. Identify the problem to be investigated
2. Find the possible methods to solve the identified problem
   a. Identify the advantages and limits of each methods
   b. Is it the only solution?
3. Identify the resources you can afford for this problem
   a. Time to learn the method
   b. Materials/instruments/computational power you have
   c. Using the current resources, estimate the time you need to do the task
4. Choose the appropriate method(s) and **plan** for your work
   a. "write your own manual" for you own topic
   b. Split your work into a few levels:
      i. What MUST be finished in order to survive (minimal requirements)
      ii. What can be done after meeting the minimal requirements, to improve your work
      iii. Bonus stuff that make your work looks fancy but maybe very difficult to do
   c. Prepare a "plan B" in case that you can't fulfil the minimal requirements with the original method
5. ~~Do~~ Try it. If failed: find the reason, adjust your plan and continue.

If you work on a protein…

- What you want to see from docking/MD simulation?
  - Enzyme-substrate interaction?
  - Design a new agonist? Design a new inhibitor?

→This would affect:

- Simulation method
- Simulation timescale
- System to be simulated
- Analysis to be done

Preparation

Before you preform MD simulation of protein

Be mentally prepared before you start!!!

- If you expect clicking a few buttons would give you all your results using computational chemistry tools
  - maybe you are just looking for a simple calculator
  - Don’t treat computational chemistry as a black box or like 1+1=2

- If you will be frustrated after getting over 5 error messages a week, then consider other methods.

No need to waste your time here.

  - Chemicals won’t give error messages if you have prepared a wrong solution, you can accept that but not an error message from a software?

Note the resources you can afford

- Time:
  - Project deadline
  - To learn the technique
  - To apply the skills you have learnt
  - To spend on trial-and-error and optimizing the experimental setup
  - To run the simulation
  - To analyze and interpret the results
- Hardware:
  - Better hardware → speed up
  - For MD: many codes utilise the GPU to accelerate the calculation
- Software:
  - Compatible with your computer?
  - Compatible with your OS?
  - Able to install?

Setting the research problem

- Why is your research important?
- Any questions still need an answer on this topic?
- What answer you want to get from doing research?

Planning your (computational) experiment

- Resources on hand
- Methods you choose
- List of tasks to be finished
  - Answer you can get by finishing the task
  - Preparation you need to do before doing the task
  - How to do the task
  - How to analyze your results
  - Priority of different tasks
- A rough schedule with the estimated time to finish each task

➢ The plan may change after you start working on it, but drafting it at the beginning helps to clear your mind, get the "big picture" of your project and ensure you have something minimal when you reach a deadline

Analysis to perform

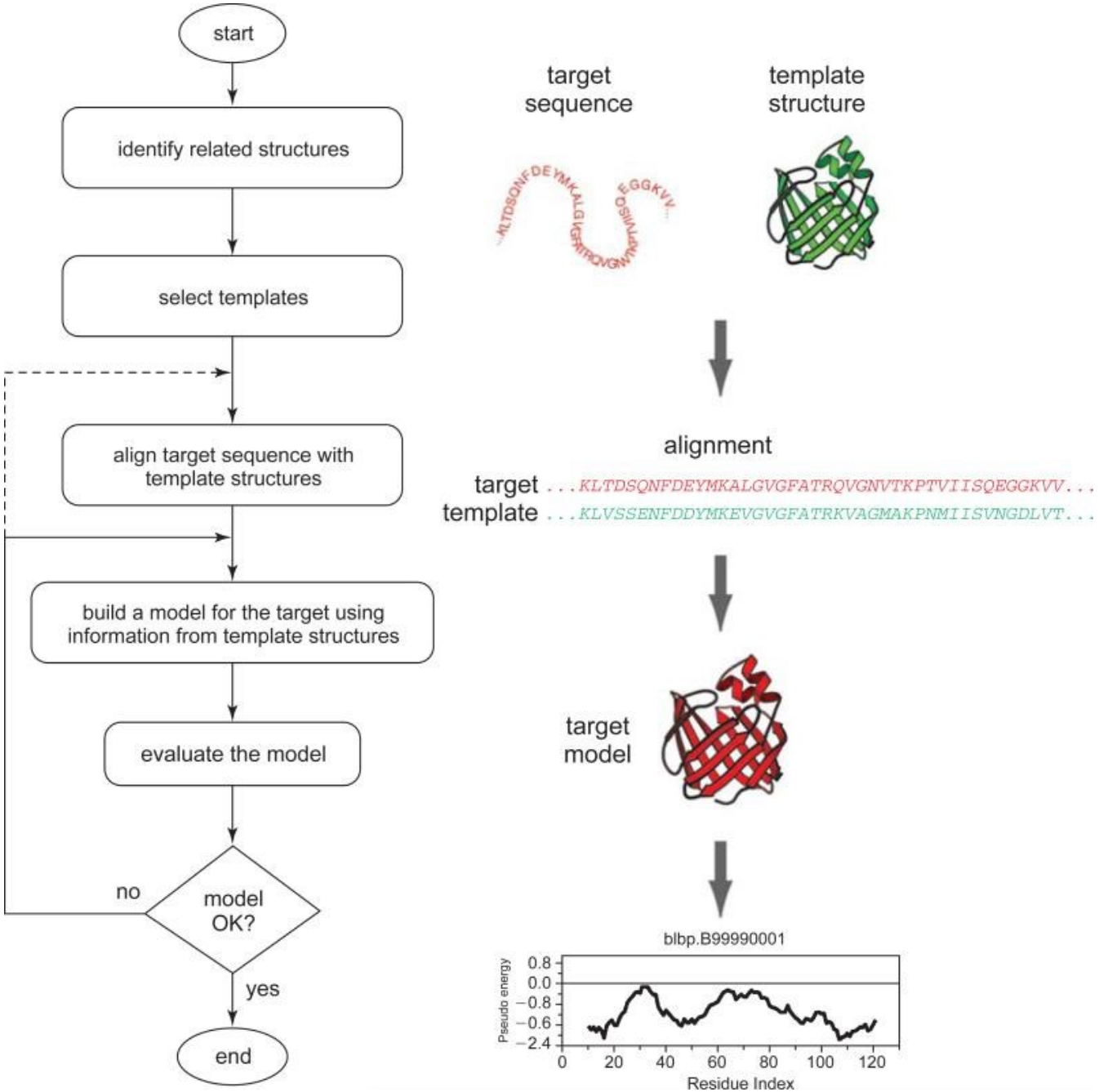
- What you need to analyze from the simulation output?
- How to interpret the raw data you get to make it something meaningful? Something that can answer your research problem?
- How to validate your results?
- Are your results reproducible?

Principles

Basic concepts of homology modelling and MD simulation

Homology Modelling

- Aka comparative modelling

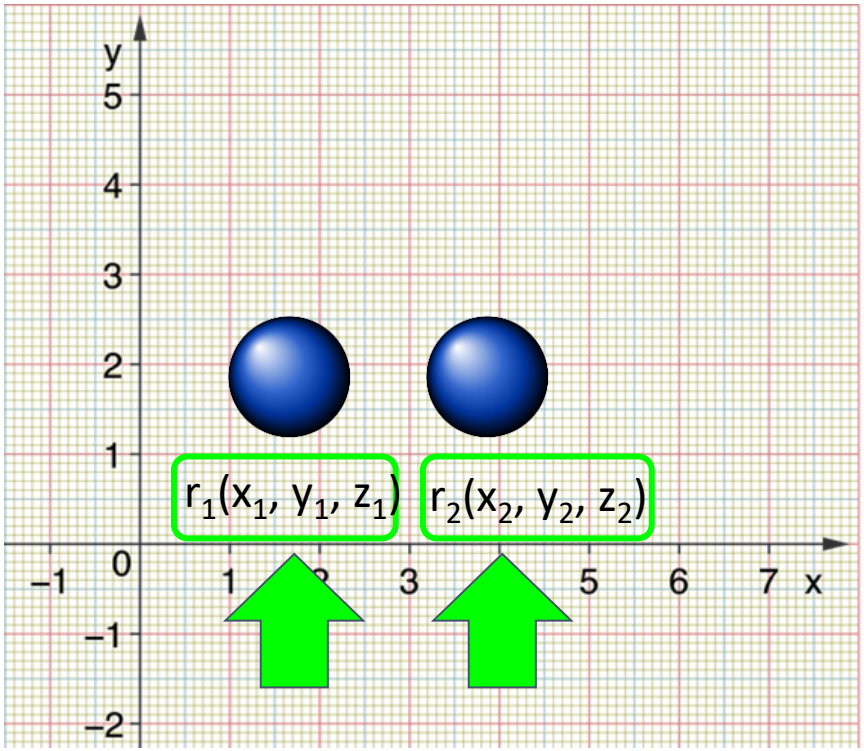


Webb, N. et al Comparative Protein Structure Modeling Using Modeller *Current Protocols in Bioinformatics* **2016**, *54,* 5.6.1-5.6.37.

Homology Modelling

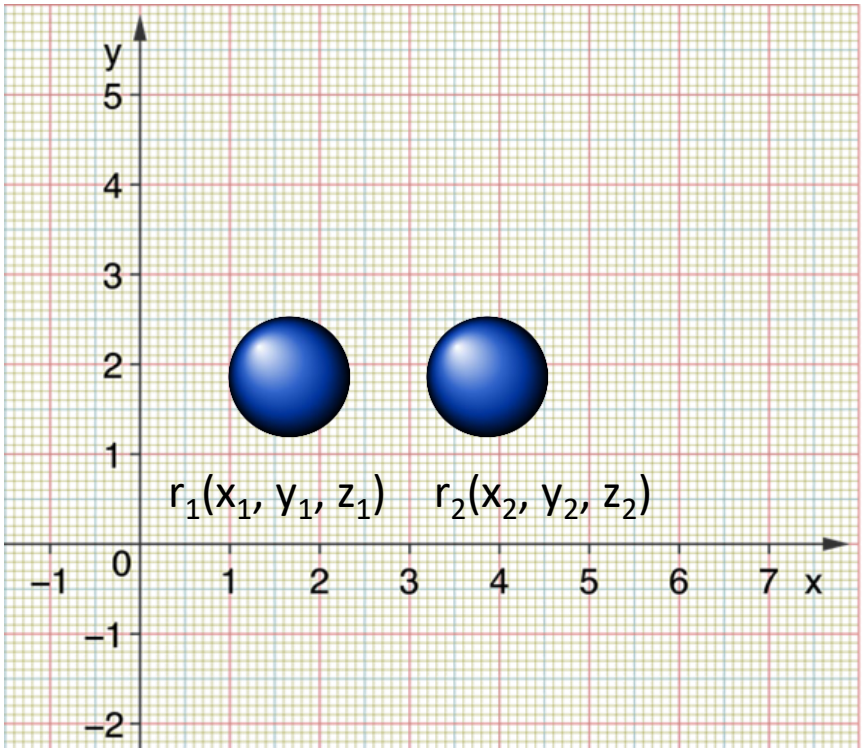
- The most import step is to validate your model!!!

Theories Molecular dynamics (MD) simulation

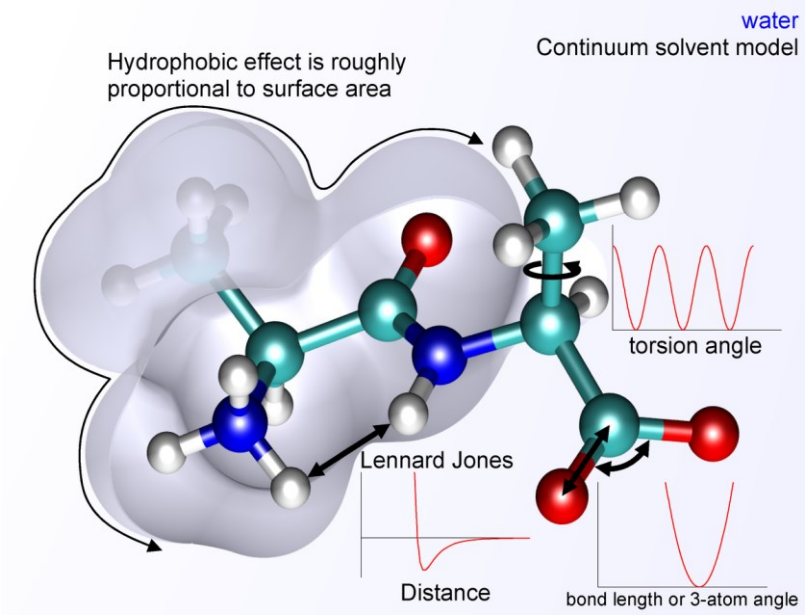


- Input initial conditions:
  - Position: r

Theories Molecular dynamics (MD) simulation

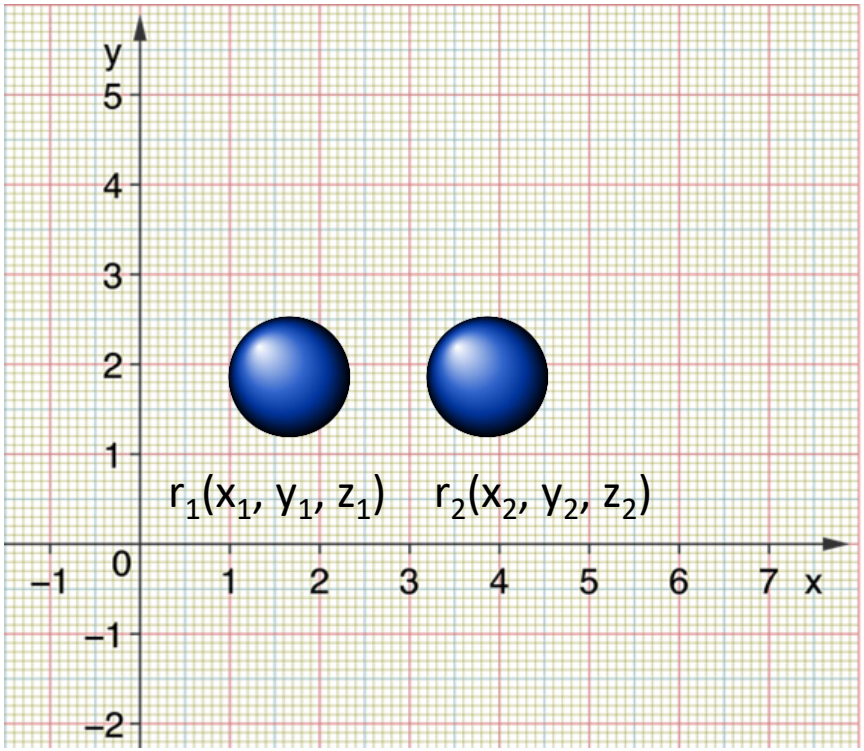


- Input initial conditions:
  - Position: r
  - Potential interaction: V(Force field)



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Theories Molecular dynamics (MD) simulation



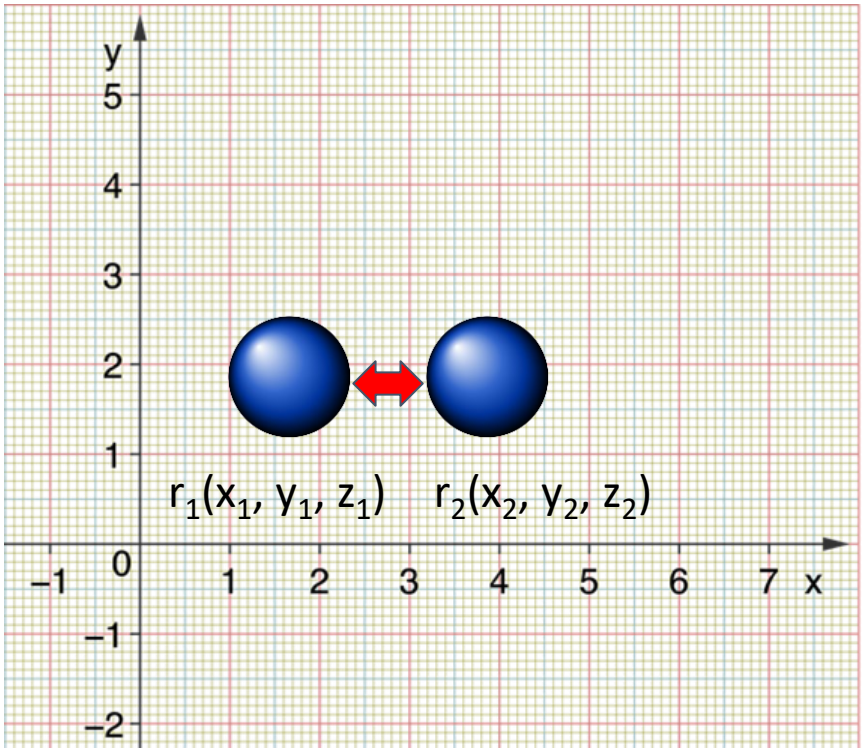
- Input initial conditions:
  - Position: r
  - Potential interaction: V(Force field)

$$E_{total} = E_{bonds} + E_{angles} + E_{dihedrals} + E_{vdw+electrostatic}$$

$$= \sum_{bonds} K_r(r - r_{eq})^2 + \sum_{angles} K_\theta(\theta - \theta_{eq})^2 +$$

$$\sum_{dihedrals} \frac{V_n}{2}[1 + \cos(n\phi - \gamma)] + \sum_{i<j} [\frac{A_{ij}}{R_{ij}^{12}} + \frac{B_{ij}}{R_{ij}^{6}} + \frac{q_i q_j}{\epsilon R_{ij}}]$$

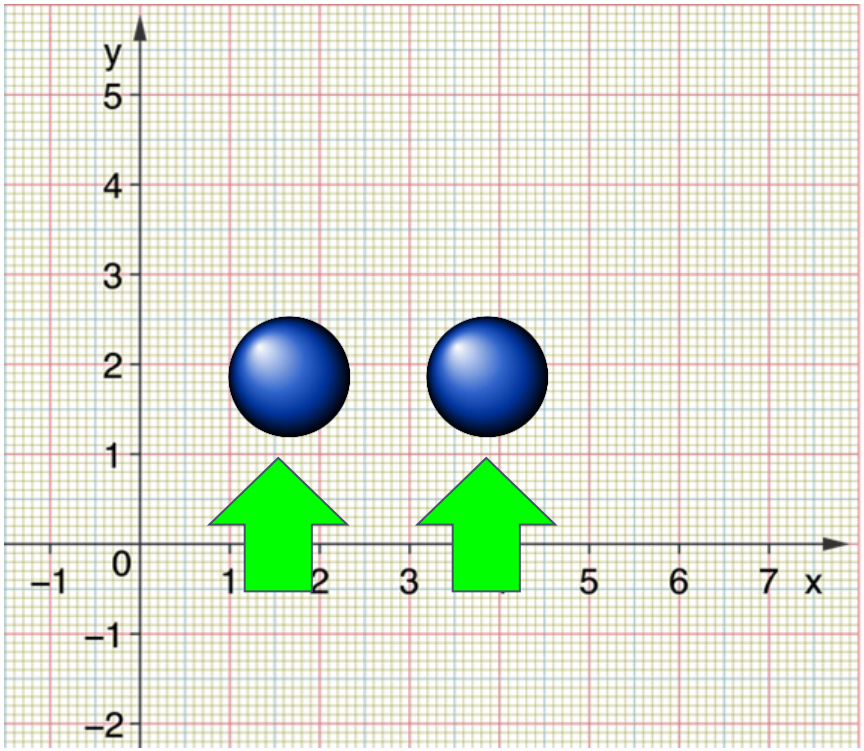
Theories Molecular dynamics (MD) simulation



- Input initial conditions:
  - Position: r
  - Potential interaction: V (Force field)
- Calculate forces: Newton's 2nd law

$$F = ma = m\frac{\partial^2 r}{\partial t^2} = -\frac{\partial V}{\partial r}$$

Theories Molecular dynamics (MD) simulation

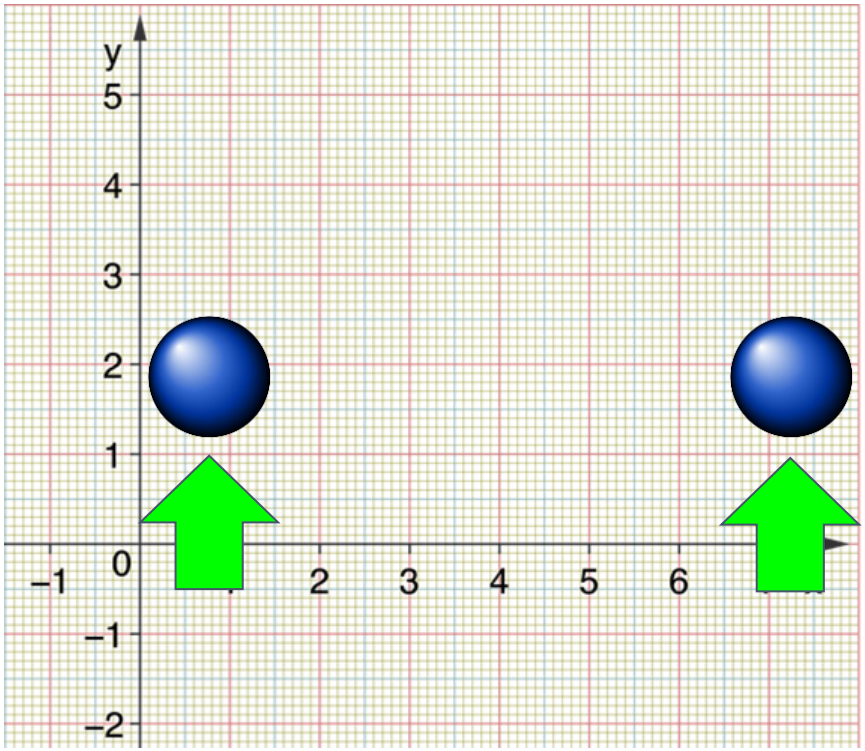


- Input initial conditions:
  - Position: r
  - Potential interaction: V(Force field)
- Calculate forces: Newton's 2nd law

$$F = ma = m\frac{\partial^2 r}{\partial t^2} = -\frac{\partial V}{\partial r}$$

- Update configurations

Theories Molecular dynamics (MD) simulation



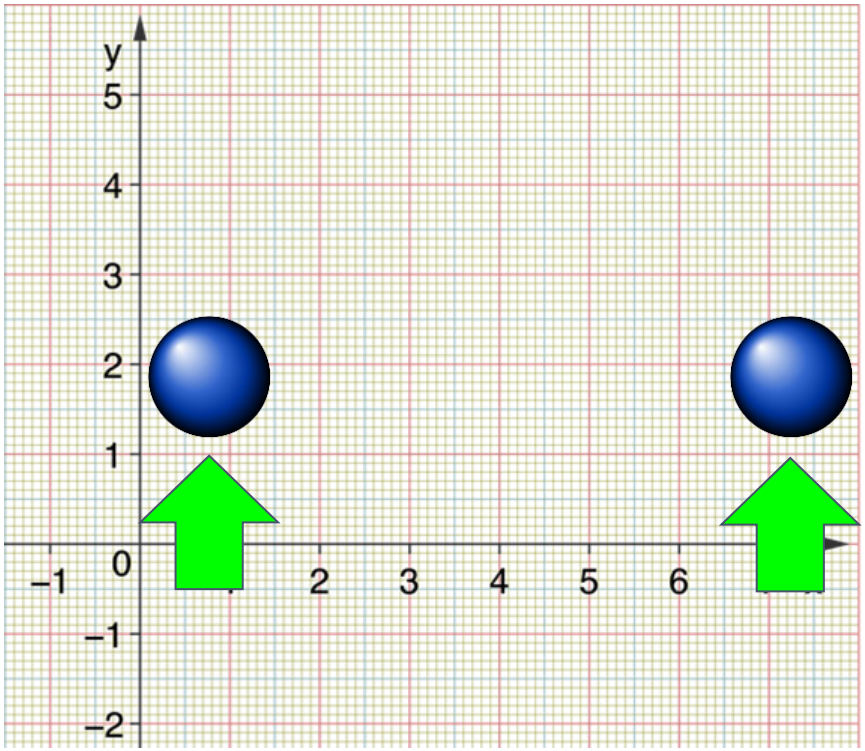
- Input initial conditions:
  - Position: r
  - Potential interaction: V(Force field)
- Calculate forces: Newton's 2nd law

$$F = ma = m\frac{\partial^2 r}{\partial t^2} = -\frac{\partial V}{\partial r}$$

- Update configurations

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = \frac{F}{m}$$

Theories Molecular dynamics (MD) simulation



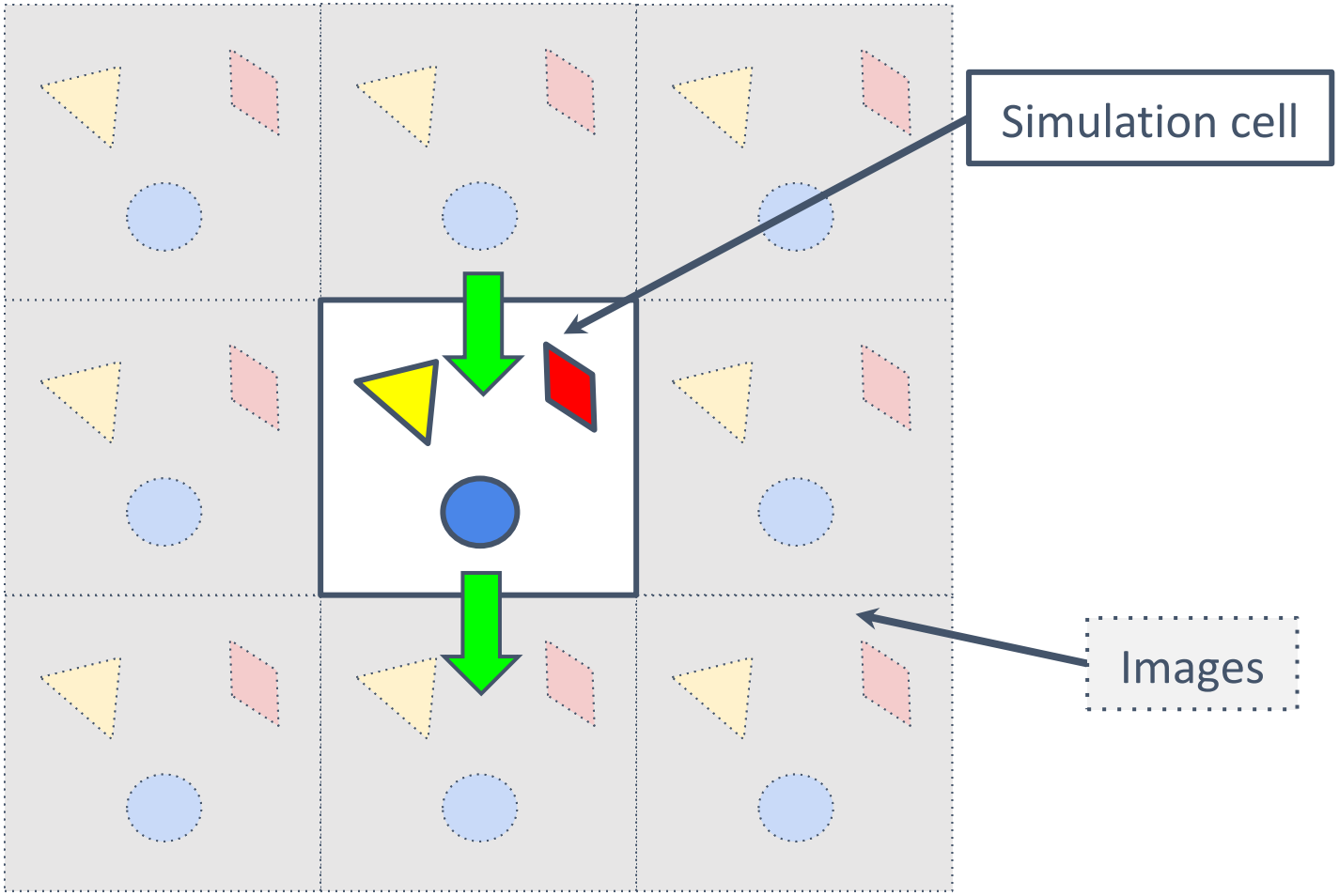
- Input initial conditions:
  - Position: r
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- Calculate forces: Newton's 2nd law

$$F = ma = m\frac{\partial^2 r}{\partial t^2} = -\frac{\partial V}{\partial r}$$

- Update configurations

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = \frac{F}{m}$$

Theories Periodic Boundary Condition (PBC)



Other parameters to consider

- Temperature coupling
- Pressure coupling
- Simulation protocol
- …

Capability of protein MD

- Study of inhibition mechanism
- Study of substrate recognition mechanism
- Stability of biomolecules in different conditions (e.g. temperature, solvent, pH, etc)
- Identify the key residues to a certain interaction
- Investigate the effect of a mutant of protein to its function
- Study of protein-xxx interactions, where xxx can be ligand/protein/DNA/RNA/membrane/nanoparticles etc.

- May or may not be possible within a short simulation timescale
  - Conformation change of protein***
  - Protein folding***
  - Binding/unbinding kinetics of a ligand***

Limits of MD that you need to be aware of

- Traditional forcefield method cannot simulate chemical reactions
- Simulation timescale of traditional MD is usually only possible within nanosecond scale with the current computational power
- Note the accuracy and conserved properties of a chosen parameter
- Energy barriers exist that trap the system in local minimum and greatly reduce the sampling efficiency
- …

Advanced techniques

- Non-equilibrium MD
  - Umbrella sampling
  - …
- Enhanced sampling methods
  - Accelerated MD
  - …
- Multiscale simulations
  - QM/MM
  - Coarse-grained MD

Practical Example

Software

- UCSF Chimera (Molecular graphics and GUI of MODELLER)
  - https://www.cgl.ucsf.edu/chimera/download.html
- MODELLER (homology modelling)
  - https://salilab.org/modeller/download_installation.html
  - https://salilab.org/modeller/registration.html
  - Use your hkbu email to request a license
- MD software/related tools: (compiled and bundled in a zip archive)
  - GROMACS (version 2019.6)
    - Open-source MD simulation software
  - QtGrace
    - Data visualization, i.e. plotting graph
- Other useful tools
  - ~~AmberTools (version 17)~~
    - ~~Input generation, analysis, also with a slower MD engine~~
  - ~~ACPYPE (software package/server)~~
    - ~~Port of more recent AMBERFF to GROMACS~~
    - ~~automatic small molecule parameterization using GAFF for GROMACS~~
  - ~~CHARMM-GUI (server)~~
    - ~~Convenient tool for generating inputs with CHARMM force field~~
    - ~~Can be problematic if you are not familiar with the file formats~~

Homology model of hen egg-white lysozyme

1. Download the sequence from uniport
   - https://www.uniprot.org/uniprot/P00698
   - [select interested region and copy to a new file]
2. Open the sequence file in UCSF Chimera
3. Search for crystal structures with matching sequence with BLAST
   - Check the E-value, resolution, presence of ligands etc
4. Load the selected structure + show in MAV
   - Remove unwanted part of the sequence before you proceed
   - Reload modified sequence (redo blast if you want), add structure sequence to MAV
5. In the MAV: open the Modeller (homology) interface
   - You may want to change the options, but for demo just use default
   - Choose the temporary directory if you want
6. Inspect the resulting model. You may choose the model with the lowest zDope score

Validation of homology model

- Web services for homology model validation
  - https://swissmodel.expasy.org/assess
  - https://servicesn.mbi.ucla.edu/SAVES
  - https://saves.mbi.ucla.edu/
  - https://prosa.services.came.sbg.ac.at/prosa.php

- Checking the homology model with MD simulation
  - Stability
  - RMSD to crystal structure (if applicable)
  - Selected system behavior matching experimental observation?
    - E.g. affinity of known inhibitors

MD simulation

- Simple & short MD of the lysozyme homology model in water
- Protocol slightly modified from Justin Lemkul's lysozyme tutorial
- Modified commands to be run on windows

Reference & Further Readings

Reference

- Software papers
  - MODELLER
    - Webb, N.; Sali, A. Comparative Protein Structure Modeling Using Modeller *Current Protocols in Bioinformatics* **2016**, *54,* 5.6.1-5.6.37.
    - Sali, A.; Blundell T.L. Comparative protein modelling by satisfaction of spatial restraints*. J. Mol. Biol.* **1993**, *234*, 779-815.
  - GROMACS
    - Abraham, M. J.; Murtola, T.; Schulz, R.; Páll, S.; Smith, J. C.; Hess, B.; Lindahl, E. GROMACS: High Performance Molecular Simulations through Multi-Level Parallelism from Laptops to Supercomputers. *SoftwareX* **2015**, *1-2*, 19–25.
  - AmberTools
    - D.A. Case, K. Belfon, I.Y. Ben-Shalom, S.R. Brozell, D.S. Cerutti, T.E. Cheatham, III, V.W.D. Cruzeiro, T.A. Darden, R.E. Duke, G. Giambasu, M.K. Gilson, H. Gohlke, A.W. Goetz, R. Harris, S. Izadi, S.A. Izmailov, K. Kasavajhala, A. Kovalenko, R. Krasny, T. Kurtzman, T.S. Lee, S. LeGrand, P. Li, C. Lin, J. Liu, T. Luchko, R. Luo, V. Man, K.M. Merz, Y. Miao, O. Mikhailovskii, G. Monard, H. Nguyen, A. Onufriev, F.Pan, S. Pantano, R. Qi, D.R. Roe, A. Roitberg, C. Sagui, S. Schott-Verdugo, J. Shen, C.L. Simmerling, N.R.Skrynnikov, J. Smith, J. Swails, R.C. Walker, J. Wang, L. Wilson, R.M. Wolf, X. Wu, Y. Xiong, Y. Xue, D.M. York and P.A. Kollman (2020), AMBER 2020, University of California, San Francisco.
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  - CHARMM-GUI
    - Jo, S.; Kim, T.; Iyer, V. G.; Im, W. CHARMM-GUI: A Web-Based Graphical User Interface for CHARMM. *J. Comput. Chem.* **2008**, *29* (11), 1859–1865.
    - Lee, J.; Cheng, X.; Swails, J. M.; Yeom, M. S.; Eastman, P. K.; Lemkul, J. A.; Wei, S.; Buckner, J.; Jeong, J. C.; Qi, Y.; Jo, S.; Pande, V. S.; Case, D. A.; Brooks, C. L., III; MacKerell, A. D., Jr.; Klauda, J. B.; Im, W. CHARMM-GUI Input Generator for NAMD, GROMACS, AMBER, OpenMM, and CHARMM/OpenMM Simulations Using the CHARMM36 Additive Force Field. *J. Chem. Theory Comput.* **2016**, *12* (1), 405–413.

Reference

- Online tutorials
  - Gromacs tutorials by Dr Justin Lemkul http://www.mdtutorials.com/gmx
  - Homology modelling tutorial (use other software) https://osf.io/82n73/wiki/Homology%20Modeling%20and%20Validation/

Further Readings

- Best practice of MD simulation
  - https://doi.org/10.33011/livecoms.1.1.5957